

Free-space dissemination of time and frequency with 10^{-19} instability over 113 km

<https://doi.org/10.1038/s41586-022-05228-5>

Received: 25 February 2022

Accepted: 11 August 2022

Published online: 05 October 2022

 Check for updates

Qi Shen^{1,2,3,9}, Jian-Yu Guan^{1,2,3,9}, Ji-Gang Ren^{1,2,3,9}, Ting Zeng^{1,2,3}, Lei Hou^{1,2,3}, Min Li^{1,2,3}, Yuan Cao^{1,2,3}, Jin-Jian Han^{1,2,3}, Meng-Zhe Lian^{1,2,3}, Yan-Wei Chen^{1,2,3}, Xin-Xin Peng^{1,2,3}, Shao-Mao Wang^{1,2,3}, Dan-Yang Zhu^{1,2,3}, Xi-Ping Shi⁴, Zheng-Guo Wang^{1,2,3}, Ye Li⁴, Wei-Yue Liu⁴, Ge-Sheng Pan^{1,2,3}, Yong Wang⁵, Zhao-Hui Li⁶, Jin-Cai Wu^{3,6}, Yan-Yan Zhang⁷, Fa-Xi Chen⁸, Chao-Yang Lu^{1,2,3}, Sheng-Kai Liao^{1,2,3}, Juan Yin^{1,2,3}, Jian-Jun Jia^{3,6}, Cheng-Zhi Peng^{1,2,3}, Hai-Feng Jiang^{1,2,3}✉, Qiang Zhang^{1,2,3,6,8}✉ & Jian-Wei Pan^{1,2,3}✉

Networks of optical clocks find applications in precise navigation^{1,2}, in efforts to redefine the fundamental unit of the ‘second’^{3–6} and in gravitational tests⁷. As the frequency instability for state-of-the-art optical clocks has reached the 10^{-19} level^{8,9}, the vision of a global-scale optical network that achieves comparable performances requires the dissemination of time and frequency over a long-distance free-space link with a similar instability of 10^{-19} . However, previous attempts at free-space dissemination of time and frequency at high precision did not extend beyond dozens of kilometres^{10,11}. Here we report time–frequency dissemination with an offset of $6.3 \times 10^{-20} \pm 3.4 \times 10^{-19}$ and an instability of less than 4×10^{-19} at 10,000 s through a free-space link of 113 km. Key technologies essential to this achievement include the deployment of high-power frequency combs, high-stability and high-efficiency optical transceiver systems and efficient linear optical sampling. We observe that the stability we have reached is retained for channel losses up to 89 dB. The technique we report can not only be directly used in ground-based applications, but could also lay the groundwork for future satellite time–frequency dissemination.

The time is one of the seven basic units of the International System of Units (SI). Since April 2020, five other basic units have also been relying on the unit of time and frequency because of the precision and stability it can achieve³. Indeed, state-of-the-art optical clocks have demonstrated the most accurate frequency reference, up to a local level of 10^{-19} (refs. ^{8,9}). To remotely access the reference optical clocks and establish a global-scale network, it is also necessary to disseminate time–frequency over long distances with an accuracy of a similar level of 10^{-19} . It is expected that an optical clock network connected by dissemination technology will open up the possibility of many exciting applications, including the next-generation definition of the unit of the ‘second’, the testing of general relativity^{7,12–14}, the probing of the changes of physical constants¹⁵, the search for gravitational waves and dark matter^{7,12,16}, and long-range quantum networks¹⁷.

So far, optical carrier-phase transfer through fibre links has reached over 1,000 km with a stability of 10^{-19} at 100 s (refs. ^{18–20}). However, such a method can make it difficult to reach certain locations such as mountains²¹, marine environments, intercontinental and interstellar ranges. In particular, to establish a global-scale network, it is essential to develop free-space time–frequency dissemination. To this end, several pioneering works have been undertaken^{10,22–26}, including those presenting many sophisticated techniques such as real-time

synchronization¹⁰, Doppler velocity correction²⁷ and carrier-phase operation²⁸. Previous work, however, has only reached a distance of up to 16 km (refs. ^{23,24}), which limits applications in many long-distance scenarios. Meanwhile, current technology cannot meet the high demands of link loss required by future satellite–ground time–frequency dissemination.

Compared with fibre links, the free-space channels in the troposphere region are much more unstable because of atmospheric disturbance, which can lead to frequent signal dropouts. Therefore, the dissemination system needs a large ambiguous range to enable link reconnection without accumulated error. The continuous-wave (CW) laser carrier used in time–frequency links²⁹ typically has an ambiguous range of a few femtoseconds, which is too small to avoid cycle slips after signal dropouts. The pulsed laser-based ranging methods, including time transfer by laser link (T2L2)³⁰ and European laser timing (ELT)³¹, have a sufficiently long ambiguous range^{30,32}; however, amplitude detection with a photodiode or a single-photon detector usually limits the instability of the link to the picosecond level. To overcome these challenges, we develop here an optical frequency comb (OFC)-based link²², which combines features of high-precision optical phase detection and a large ambiguous range. The linear optical sampling (LOS) offers femtosecond accuracy over the whole ambiguous range^{22,24}. By using 1 W

¹Hefei National Research Center for Physical Sciences at the Microscale and School of Physical Sciences, University of Science and Technology of China, Hefei, China. ²Shanghai Research Center for Quantum Science and CAS Center for Excellence in Quantum Information and Quantum Physics, University of Science and Technology of China, Shanghai, China. ³Hefei National Laboratory, University of Science and Technology of China, Hefei, China. ⁴Faculty of Information Science and Engineering, Ningbo University, Ningbo, China. ⁵Xinjiang Astronomical Observatory, Chinese Academy of Sciences, Urumqi, China. ⁶Key Laboratory of Space Active Opto-Electronic Technology, Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai, China. ⁷Key Laboratory of Time and Frequency Primary Standards, National Time Service Center, Chinese Academy of Sciences, Xi'an, China. ⁸Jinan Institute of Quantum Technology, Jinan, China. ⁹These authors contributed equally: Qi Shen, Jian-Yu Guan, Ji-Gang Ren. ✉e-mail: hjiang1@ustc.edu.cn; qiangzh@ustc.edu.cn; pan@ustc.edu.cn

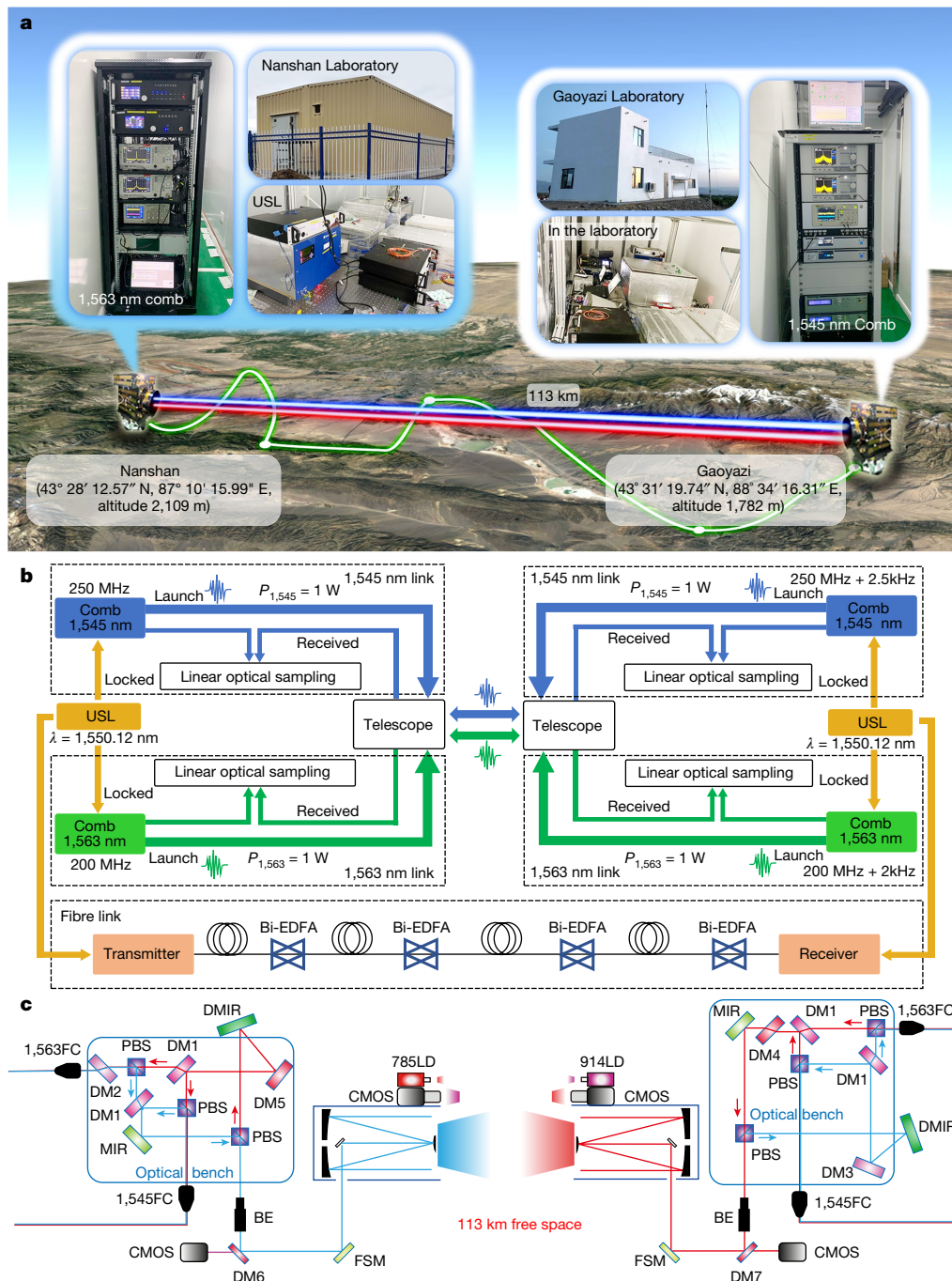


Fig. 1 | The experimental setup. a, Overview of the 113 km free-space time-frequency dissemination experiment. **b**, The main diagram of the system, which contains a 1,545 nm free-space link, a 1,563 nm free-space link and a fibre link. $P_{1,545}$ ($P_{1,563}$), output power of the 1,545 (1,563)-nm comb. **c**, The optical layout of the optical transceiver telescope, where different colours of optical paths indicate different polarizations. PBS, polarizing beam splitter; 1,545FC

(1,563FC), fibre collimator for two-way 1,545 (1,563) nm comb transceiver; DM, dichroic beam splitter; MIR, mirror; DMIR, deformable mirror; WDM, wavelength division multiplexer; BE, beam expander; FSM, fast steering mirror; CMOS, complementary metal-oxide semiconductor; 785LD (914LD), 785-nm (914-nm) laser diode. The underlying map in **a** is from Google, DigitalGlobe.

OFCs and nanowatt-level LOS modules, we achieved a time-frequency dissemination over a free-space link of 113 km and obtained a stability of below 4×10^{-19} at 10,000 s.

The experiment was performed in Urumqi, Xinjiang Province. As shown in Fig. 1, two terminals (A and B) are located at Nanshan and Gaoyazi with a distance between them of 113 km. Each terminal is equipped with an ultra-stable laser (USL), two high-power OFCs with different wavelengths centred at 1,545 nm and 1,563 nm, two LOS modules and an optical transceiver telescope. An USL with a 3×10^{-15}

frequency instability at 1 s at a wavelength (λ) of 1,550.12 nm is used as the reference clock source. The OFC optical phase locked to the USL is used as the carrier and reference signals of the local sampling (Methods). By frequency multiplexing the common free-space channel, we establish two independent two-way time-frequency transfer links, enabling precise evaluation of the link performance without limitation from the USL. As the two multiplexing channels share the same free-space link, common-mode noise occurs. To better evaluate our system, we also established an independent fibre link connecting

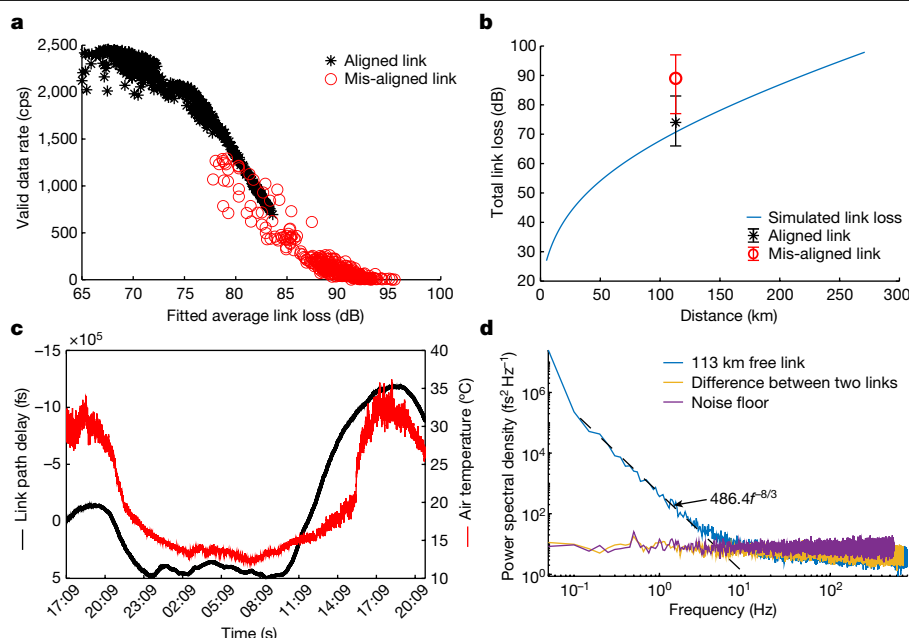


Fig. 2 | Characterization of the 113 km free-space link. **a**, Relation between the link loss and the valid data rates of the 1,545 nm link. **b**, Link loss at different distances. The error bars of the aligned and mis-aligned links are the differences between the minimum and maximum values and the mean values.

c, Optical link path delay measured by the 1,545 nm link and the air temperature measured at terminal B. **d**, Power spectral densities of the free-running and compensated link delay. The noise floor is measured by connecting two terminals with a 10 m fibre.

the two terminals with a distance of 209 km (Methods). All links (so the 1,545 nm free-space, 1,563 nm free-space and fibre links) share the same USL at each terminal.

The repetition rate of the 1,545 nm OFC is 250 MHz and the differential repetition rate between two terminals is set to 2.5 kHz, whereas the repetition rate of the 1,563 nm OFC is 200 MHz and the differential repetition rate is set to 2 kHz. To overcome the high loss of the link, two-stage high-power erbium-doped fibre amplifiers (EDFAs) are used, providing a 1 W output power after filtering with a 20 nm spectrum bandwidth, which is approximately 5 to 10 times higher than the OFC power in the previous experiments. The 3-dB bandwidth of both the 1,545 and 1,563 nm OFCs is approximately 7 nm, which is determined by the gain region of the EDFA. A short pulse corresponds to low noise and high signal-to-noise interference; however, high peak power can induce unexpected nonlinearity and damage. Considering that the systems have to continuously operate over long time periods, we produce a chirped pulse to reduce the power density to avoid burns. The pulse widths at the output are between 60 and 90 ps for the 1,545 and 1,563 nm OFCs (Supplementary Information). Water coolers are used to protect the EDFAs from overheating; in addition, they reduce the temperature fluctuations by a factor of approximately ten.

The optical part of the LOS (Fig. 1b) is made up of pig-tailed optical components housed in an aluminium box. To minimize the thermal drift effect on the non-reciprocal fibre path, the length of the mismatched fibres is optimized to be below 5 cm, and the temperature is stabilized with a thermoelectric cooler, which has an in-loop deviation of 7 mK corresponding to a time deviation below 0.03 fs. The interference data between the local and received OFC signals are detected by using low-noise balanced detectors, digitized with analog-to-digital converters, and recorded with a field-programmable gate array to determine the arrival time of the laser pulse. All electronics share the same reference clock signal of the repetition rate of the local comb. Owing to the global positioning system receivers at both terminals, the start time of the data collection for both terminals can be synchronized to be within 30 ns. The LOS can determine the arrival time of the received OFC according to a single overlapping process between two OFCs.

However, only if the recorded data show a clear interaction peak can the arrival time be determined precisely.

Two dedicated optical transceiver telescopes with automatic direction tracking functions were developed for this experiment. The telescopes in the two laboratory terminals are equipped with a beacon laser for direction tracking at wavelengths of 785 nm and 914 nm, respectively. The primary mirror of each telescope is a Cassegrain reflector with an aperture of 400 mm and a focal length of 1,600 mm. We use an orthogonal polarization scheme to separate the sending and receiving OFCs. The orthogonal polarization setting is designed to provide large isolation of two-way transfer, which is also used to simulate a real satellite-ground scenario in which the relative angular motion separates the backwards and forwards transfers^{33,34}. Such a design induces a non-reciprocal optical path. To minimize instability induced by the non-reciprocal paths, we bonded all of the optical components onto a plate of fused silica, and the temperature was stabilized to be approximately 1 °C peak-to-peak. A deformable mirror was inserted in the receiving optical path to manipulate the adaptive optics and compensate for incoming wavefront distortions caused by the atmosphere, therefore improving the coupling efficiency from free space into a single-mode fibre.

The key point of performing the experiment over a long link is to overcome large link loss, so that recording this link loss is important to characterize the condition of the link. However, the stray reflection of the local OFC is usually much stronger than the received OFC; thus, it cannot be measured directly. To monitor the link loss, we developed an approach of determining the link loss according to the peak voltages of the LOS, the valid data rates and the statistical loss distribution (Supplementary Information). The accuracy of this approach is calibrated by using a commercial optical power meter. Figure 2a shows the relation between the link loss and the valid data rates (Sn, amount per second). Each point is an average value measured in 160 s. Typically, the loss of the aligned 113 km link fluctuates between 66 and 83 dB, with an average value of 74 dB. The maximum link loss of 83 dB corresponds to a valid data rate of more than 600 counts per second (cps), that is, about 25% of the total. To evaluate the loss tolerance, we deliberately

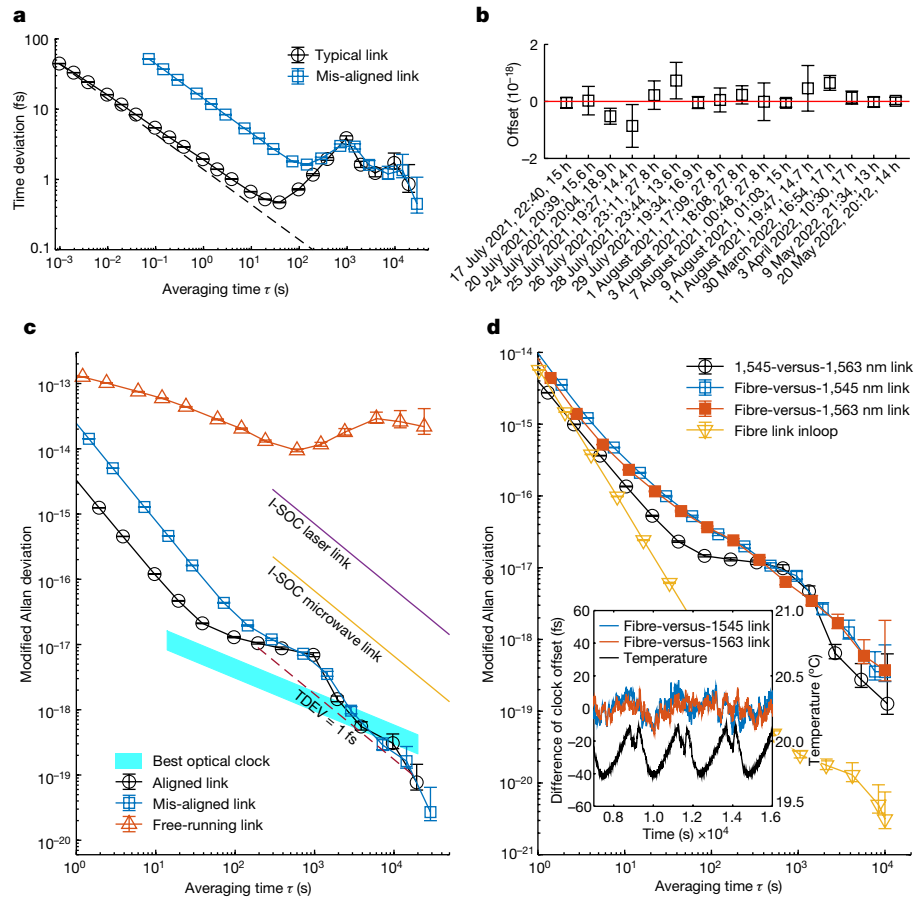


Fig. 3 | Experimental results of time–frequency transfer. **a**, The TDEV of one free-space link. The error bar is 1σ . **b**, The fractional offset, or bias, across 15 datasets, each labeled by start time and duration. The uncertainty per point is the MDEV value at 10,000 s. The red line is the weighted average. **c**, The fractional frequency instability of links. The error bar is 1σ . Black circles, well-aligned 113 km free-space time–frequency link; blue squares, mis-aligned link; orange triangles, free-running link. As the two links have similar performance, the instability of one link is divided by $\sqrt{2}$. The performances of the best optical clock^{39,40}, the I-SOC (Space Optical Clock on the International

Space Station) laser link, the I-SOC microwave link and the TDEV of 1 fs are also shown. Here the original data of the best optical clocks are the Allan deviation. The MDEV is $\sqrt{2}$ times smaller provided that the noise type is white frequency noise⁴¹. **d**, Time comparison among the fibre link and two free-space links. The error bar is 1σ . Black circles, two free-space links; blue squares, fibre and 1,545 nm comb link; orange squares, fibre and 1,563 nm comb link; yellow triangles, in-loop stability of the fibre link. The inset shows the trend of the difference between the fibre link and the free-space links with temperature.

mis-aligned the optical path to increase the average loss up to 89 dB. In fact, even when the average loss exceeds 90 dB, corresponding to a 1 nW average received power, valid data rates can be more than 100 cps. The highest valid data rates, close to 100% for an aligned link and up to 50% for a mis-aligned one, always occur at twilight, whereas the lowest data rates appear at noon because of the strong atmospheric vortex caused by heating by the sun. The distribution of the link loss is different for the two working conditions, whereas the dynamic ranges are almost the same. The link loss is also estimated theoretically according to our telescope and atmospheric parameters, as shown in Fig. 2b (Supplementary Information), and it is consistent with the measured link loss. The 89 dB average loss of the mis-aligned link is measured from the output of the comb to the input of the balanced detector, including around 8.8 dB for the fixed loss of the telescope optical efficiency and the LOS optics loss of the transmitter/receiver (Supplementary Information), which corresponds to the loss of a 190 km link according to the simulation.

The link delay fluctuations, which can be obtained from timing data, fundamentally limit the link instability because the link delay results limit noise cancellation. Figure 2c shows that the link delay drifts more than 1 ns peak-to-peak over 100,000 s (about 28 h), and this is attributed to temperature fluctuations. We choose the 200 s link path delay data

of Fig. 2c to further analyse atmospheric noise in the frequency domain, giving the noise power spectral density (Fig. 2d). At low frequencies below 3 Hz, the power spectral density of the free link shows a single power-law decay at $f^{-8/3}$, where f is the Fourier frequency, which is dominated by the turbulence-induced atmospheric piston effect, and is consistent with the proposed atmospheric model^{22,35,36}. The two-way time transfer further suppresses the link noise by a factor of $|1 - \exp(i2\pi\tau f)|^2$, where τ is the link delay. The time jitter of the suppressed phase noise from 2.6 kHz (sampling rate) to 10 MHz is only 32 attoseconds and it is therefore negligible. As a result, the link noise is the same as the noise floor measured by a very-short-fibre link, as shown in Fig. 2d. The noise floor is attributed to downconverted high-frequency noise related to the laser pulse width, sampling rates and dispersion management.

Having fully characterized the 113 km link, we carried out a time–frequency transfer experiment and obtained a total of 15 datasets. The continuous acquisition duration of each dataset was between 13 and 27.8 h. The link can operate almost all the time, except during periods of rain or at noon on sunny days. At first, the instability of the link is evaluated by comparing two wavelength multiplexing links, and we assume that the two links have similar performance. Figure 3a shows the time deviation (TDEV) of the link. It descends with a slope of $\sqrt{\tau}$ for short averaging times (τ) and reaches 0.5 fs at 40 s. The starting

points for both TDEVs are around 40 fs, because the data are obtained only if the interference signal is over a threshold voltage, corresponding to an instantaneous power of 2–4 nW. For long averaging times, the thermal drift effect dominates the TDEV, which is limited to around 1 fs. The peaks at around 1,000 s correspond to the control cycle of the air conditioners. Note that the thermal drift effect can be further improved by passively matching the optical path, and by thermal stabilization or active stabilization of the optical phase.

The frequency transfer performance is presented by the modified Allan deviation (MDEV), as shown in Fig. 3c. The relative instability is below 3×10^{-19} at 10,000 s and in the range of 10^{-20} at 20,000 s. Even for the mis-aligned link, the instability reaches 10^{-19} at a few thousand seconds. Compared with the best atomic clock, the link is more stable for an integration time longer than 1 h. The link instability is about $7 \times 10^{-14}/(\tau^{3/2} \times S_n^{1/2})$ determined by the random noise of the laser system and the sampling process for short averaging times, whereas it is a few $10^{-15}/\tau$, being limited by the thermal effect, for long averaging times. The instability is more than three orders of magnitude smaller than that of commonly used microwave links^{37,38} and more than one order of magnitude smaller than that of underdeveloped multichannel microwave links and laser links³¹. Instabilities for short averaging times are worse than those found in demonstrations of other comb-based transfers^{10,22,23}, mainly because of the use of high-power amplified combs. To avoid nonlinear effects, we used highly chirped laser pulses, which involves a large dispersion difference. However, this leads to poor accuracy of the time determination. In principle, this effect can be minimized by careful dispersion management.

The systematic fractional offset is another important parameter for evaluating the performance of the time–frequency transfer link in addition to the stability, and it can be obtained by calculating the non-zero slope of the measured link residual time offsets. The fractional offset shown in Fig. 3b is 3.4×10^{-19} (1 σ) with a weighted average of 6.3×10^{-20} . The performance of such a link meets the requirements of an upcoming new SI metrology system. A mandatory step of SI second redefinition is to compare different atomic clocks with instabilities below 5×10^{-18} , which can be done in 1 h.

Relative instabilities among two free-space links and the fibre link are in the range of $1\text{--}4 \times 10^{-19}$ at 10,000 s as shown in Fig. 3d. Thermal drift effects (see the inset in Fig. 3d) limit the link performance for long averaging times. Although in-loop instability of the fibre link is negligible, there is about half a metre of out-of-loop fibre that is used to connect the fibre link at each terminal. The stability of the two space links is slightly better because the location of the equipment of the two free-space links is closer than that of the fibre link. In fact, the phase noise of the atmosphere is much less than that of the fibre link and has been efficiently suppressed by the two-way operation (Fig. 3d). So, the thermal drift effect should be concentrated in the fibre parts. Furthermore, we also implement real-time synchronization with our system (Supplementary Information).

In summary, we have implemented a time–frequency dissemination over a 113 km free-space link. The TDEV of the link is around 1 fs for long averaging times and the relative instability is below 4×10^{-19} at 10,000 s, whereas the link loss is up to 89 dB. Several key techniques, particularly watt-level OFCs, the separation of the receiving and transmitting light using an orthogonal polarization scheme and the high-sensitivity LOS detection, have been used and verified, and this opens the path to satellite–ground time–frequency dissemination. On the basis of these techniques, we anticipate that long-haul free-space OFC links, combined with fibre-based and satellite-based time–frequency links, will become important parts of future optical clock networks. The Doppler effect will be a challenge for satellites in the future because of the relevant transfer asymmetry. A previous study²⁷ has demonstrated a 10^{-19} instability of the link under low Doppler velocities, such as 24 m s^{-1} , and the large Doppler velocity condition still needs to be investigated further.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-022-05228-5>.

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Methods

Detailed experimental setup and parameters

The experimental setup of one terminal is shown in Extended Data Fig. 1. The setups of the two terminals are identical, and they each contain two OFCs and one USL. The USL acts as the reference clock, to which the two OFCs are phase locked. The upper part of Extended Data Fig. 1 shows the laser source of the 1,545 nm link with a wavelength of $1,545 \pm 7$ nm and the bottom part shows the laser source of the 1,563 nm link with a wavelength of $1,563 \pm 5.5$ nm. For each OFC, a high-power EDFA is used to amplify the power to about 1 W. We built a small integrated optical fibre module to produce the optical beatnotes between the OFCs and the USL, as well as the LOS. For the 1,545 nm OFC, a 1×3 beam splitter divides the beam into three parts with a ratio of 1:2:97. The 1% part is used as the local beam of the LOS. The 97% part is connected to the telescope. The last 2% part goes through a 0.8 nm filter and then beats with the USL. For the 1,563 nm OFC, the beam first goes through a 20 nm filter centred at 1,563 nm. The reflected part further passes through a 0.8 nm filter and then beats with the USL. The light passed is then divided by a 1:99 beam splitter. The 1% part is used as the local beam of the LOS and the 99% part is connected to the telescope. All non-reciprocal optical paths lie in the integrated module. The module's temperature is stabilized with a standard deviation of about 7 mK.

We used two independent analog-to-digital converters and field-programmable gate arrays to record and process the signals. Their reference clocks were synchronized with the repetition rate of the corresponding OFC. A global positioning system receiver provides a time synchronization service so that all the electronics at the two terminals can share the same starting time. This is crucial for synchronizing the acquisition data at the two terminals. In each terminal, a rubidium radio frequency reference synchronizes all the locking electronics. The carrier-envelope offset (CEO) frequency (f_{ceo}) and the beat frequency (f_{beat}) are set to be equal and in opposite directions so that they cancel each other out. Details are shown below.

When all phase-locked loops are closed, we have the relation $f_{\text{cw}} + f_{\text{beat}} = f_{\text{ceo}} + N \times f_{\text{rep}}$, where f_{cw} is the CW frequency of the USL, f_{rep} is the repetition frequency of the OFC, and N is an integer. Here we set f_{ceo} to be the same as f_{beat} , to make $f_{\text{cw}} = N \times f_{\text{rep}}$. Thus, the repetition rate of the OFC only depends on the frequency of the USL. Frequencies are listed in Extended Data Table 1. We set the repetition rate of the 1,545 nm OFC at terminal A (Nanshan) to be 250 MHz. In fact, all frequencies are determined by the frequency of the USL and the daily drift between two USLs is several kilohertz.

Optical fibre time–frequency transfer link

The total fibre length is 209 km and the loss is 65 dB. To compensate for the transmitted loss, three bidirectional amplifiers are put on the path between the terminals and one is put at terminal B. The detailed configuration is shown in Extended Data Fig. 2.

At terminal A, we split the USL into two parts. One is used to lock the combs and the other goes through an optical module for fibre time–frequency transfer. The module is a highly unbalanced Michelson interferometer, with the short arm in the local optical module and the long arm up to terminal B. The acoustic optical modulator (AOM) at terminal A is used to compensate for the noise in the fibre link, and the AOM at terminal B is used to distinguish the reflected laser from terminal B with the backward scattering signal during the link. The reference clock for the locking electronics is synchronized with the repetition rate of the comb, to ensure that the fibre time–frequency transfer system shares the same reference frequency with the free-space system. At terminal B, the local USL beats with the signal laser. The beating signal is filtered with a tracking oscillator and measured with a dead-time free-phase recorder. Direct digital synthesis converts the repetition rate of the 1,563 nm OFC to 10 MHz to drive the phase recorder.

In the free-space time–frequency transfer link, the clock offset is calculated with the time base that came from the 1,545 nm OFC. The phase difference of the fibre link should be converted to this time base too before we compare it with the free-space link. At terminal A, the USL and the locking electronics are all synchronized with the time base, so that the frequency of the signal transmitted to terminal B (f_{signal}) can be calculated. At terminal B, we also record the ratio (k) between the frequency of the USL and the 10 MHz. When the phase recorder reads the frequency as f_{beat} , the real frequency at the time base of terminal A should be $f_{\text{cw,B}}/k \times (f_{\text{beat}}/10 \text{ MHz})$. We then have

$$f_{\text{cw,B}} + f_{\text{cw,B}}/k \times (f_{\text{beat}}/10 \text{ MHz}) = f_{\text{signal}}.$$

The value of $f_{\text{cw,A}} - f_{\text{cw,B}}$ can be easily calculated from the above formula. To convert the frequency to time, one needs to be aware that the reference frequencies of the phase recorder at the different terminals are not same. Another ratio is needed to correct the gate time. This ratio can also be calculated by considering $f_{\text{cw,B}}$ and $f_{\text{cw,A}}$. Finally, the offset between the start recording time of the free-space link and the fibre link can be determined by minimizing the phase drift of the result of the comparison. Most of our fibre links are aerial optical fibres. There are several breakpoints in the fibre link data, which relate to the occurrence of optical cycle slip. For each breakpoint, we remove 50 s of data before and after this point.

Time comparisons among the fibre link and two free-space links are measured. The stability of the fibre-versus-1,545 nm link is 3.5×10^{-19} at 10,000 s and the stability of the fibre-versus-1,563 nm link is 3.7×10^{-19} at 10,000 s. The stability of the 1,545-versus-1,563 nm link is 1.3×10^{-19} at 10,000 s. There is a common-mode noise between the 1,563 and 1,545 nm link and its value should be below 4×10^{-19} at 10,000 s, which is the result of the thermal effect of the fibres as the out-of-loop connecting fibre link located outside of the original thermal shield box. This result shows that the stability of the difference between fully independent links can reach a 4×10^{-19} level. In fact, as the phase noise of the atmosphere caused by the piston effect and temperature drift is concentrated at low frequencies below 10 Hz, the phase noise is efficiently suppressed by the two-way operation. The noise of the aerial optical fibre is much larger than the noise of the atmosphere. To estimate the noise intensity of the fibre link, we record the frequency of the driver signal for AOM1, which is used to compensate for the Doppler noise in the fibre link. The relative phase drift of the link can be calculated from the frequency data (Supplementary Information). The frequency noise of the fibre link can be up to several kilohertz and the phase drift is hundreds of nanoseconds, which is almost two orders of magnitude larger than that of the free-space link.

Data availability

All data generated or analysed during this study are included in this published article (and its supplementary information files).

Code availability

All relevant codes or algorithms are available from the corresponding author upon reasonable request.

Acknowledgements This work is supported by the National Key Research and Development Programme of China (grant nos. 2017YFA0303900, 2020YFA0309800 and 2020YFC2200103); the Strategic Priority Research Programme of Chinese Academy of Sciences (grant nos. XDB35030000 and XDA15020400); the National Natural Science Foundation of China (grant nos. T2125010 and 61825505); the Anhui Initiative in Quantum Information Technologies (grant no. AHY010100); the Key R&D Plan of Shandong Province (grant nos. 2020CXGC010105 and 2021ZDPT01); the Key Research and Development Programme of Guangdong Province (grant no. 2018B030325001); the Shanghai Municipal Science and Technology Major Project (grant no. 2019SHZDZX01); and the Innovation Programme for Quantum Science and Technology (grant no. 2021ZD0300100).

Article

Author contributions H.-F.J., Q.Z. and J.-W.P. conceived the experiment. Q.S., J.-Y.G., L.H., M.L., J.-J.H., M.-Z.L., Y.-W.C., X.-X.P., H.-F.J. and Q.Z. designed the time and frequency setup. J.-G.R., Y.C., T.Z., J.-C.W., J.-J.J., J.Y. and C.-Z.P. built the optical telescopes. L.H., X.-X.P., Y.-Y.Z. and H.-F.J. developed the 1,563 nm OFCs and amplifiers. Q.S., M.L., J.-Y.G., J.-J.H., M.-Z.L. and S.-K.L. developed the LOS optics and electronics and the real-time synchronization modules. J.-Y.G., F.-X.C., H.-F.J., Q.S., J.-J.H. and M.-Z.L. developed the optical fibre transfer link. W.-Y.L., X.-P.S., Y.L., M.L., Q.S. and J.-Y.G. designed the data acquisition software of the LOS. J.-Y.G., Q.S., M.L., J.-J.H., M.-Z.L. and Y.-W.C. designed the time data process software. All authors carried out the experiment, analysed the data and contributed to the writing of the paper.

Competing interests The authors declare no competing interests.

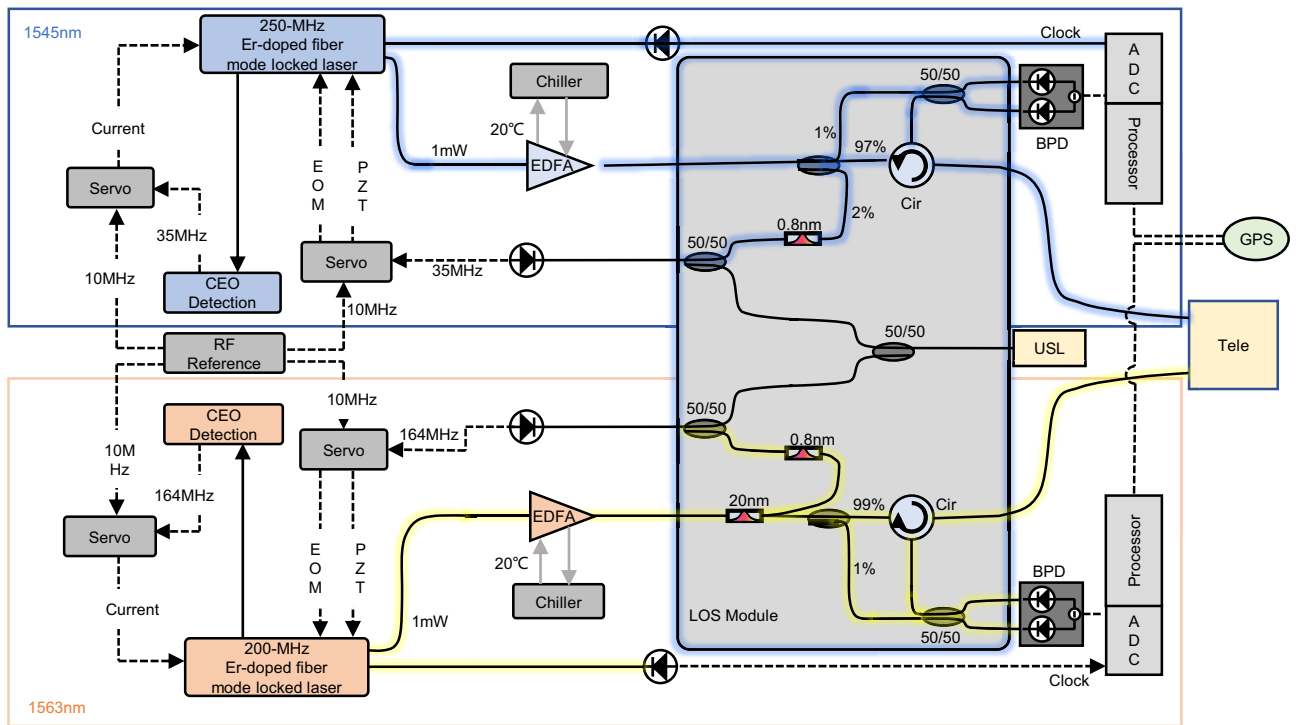
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-022-05228-5>.

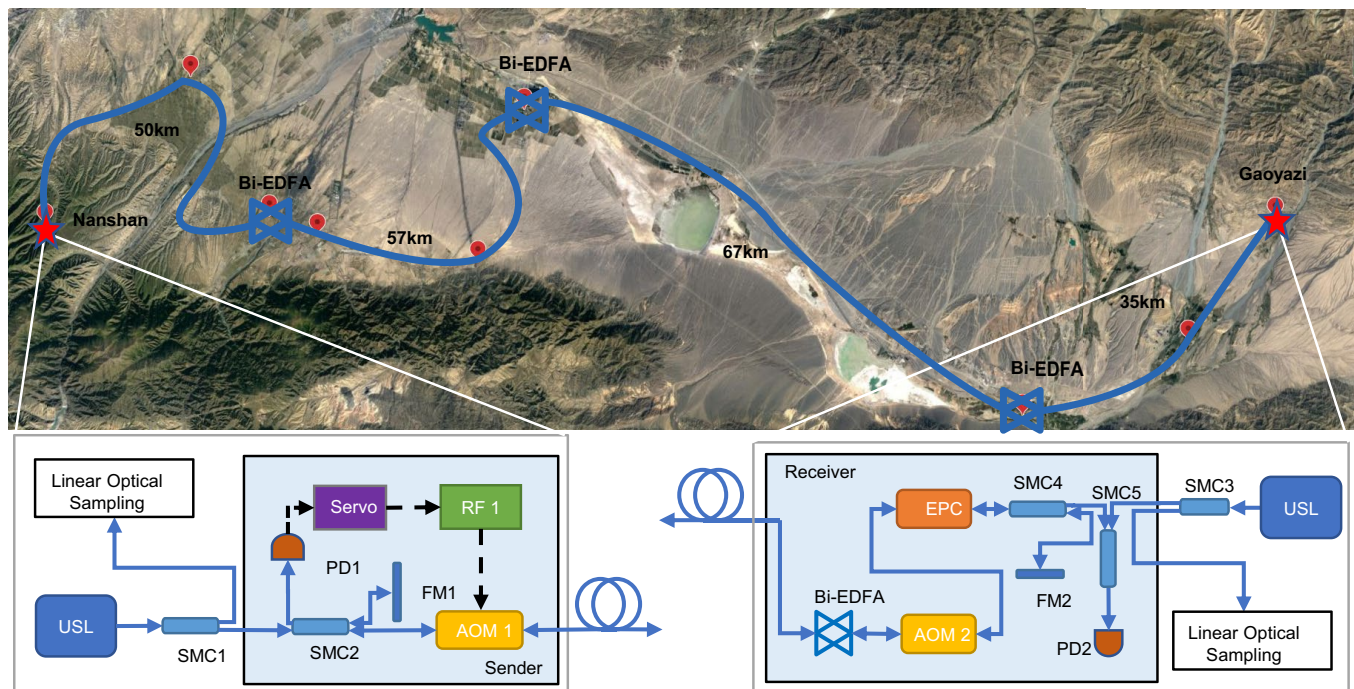
Correspondence and requests for materials should be addressed to Hai-Feng Jiang, Qiang Zhang or Jian-Wei Pan.

Peer review information *Nature* thanks the anonymous reviewers for their contribution to the peer review of this work. Peer reviewer reports are available.

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Extended Data Fig. 1 | Detailed experimental optics setup of single terminal. USL, ultra-stable laser; EDFA, erbium-doped fiber amplifiers; Cir, circulator; BPD, balanced photodiode; Tele, Telescope.



Extended Data Fig. 2 | Setup for fibre time-frequency transfer. USL, ultra-stable laser; SMC, single-mode coupler; PD, photon diode; RF, radio frequency source; FM, Faraday mirror; AOM, acoustic optical modulator;

EPC, electric polarization controller; Bi-EDFA, bidirectional erbium doped fiber amplification. Underlying map from Google, DigitalGlobe.

Extended Data Table 1 | Detailed parameters of the experiment

Parameters	Symbols	Values
Frequency of the ultra-stable laser at terminal A	$f_{cw,A}$	193.401 THz
Repetition rate of Comb 1545 at terminal A	$fr_{1545,A}$	250 MHz
Index of the tooth of Comb 1545 locking to the ultra-stable laser at terminal A	$N_{45,A}$	773604
Repetition rate of Comb 1563 at terminal A	$fr_{1563,A}$	199.99504 MHz
Index of the tooth of Comb 1563 locking to the ultra-stable laser at terminal A	$N_{63,A}$	967029
Frequency of the ultra-stable laser at terminal B	$f_{cw,B}$	$f_{cw,A} - 1.8675$ MHz
Repetition rate of Comb 1545 at terminal B	$fr_{1545,B}$	$fr_{1545,A} + 2582.9$ Hz
Index of the tooth of Comb 1545 locking to the ultra-stable laser at terminal B	$N_{45,B}$	773596
Repetition rate of Comb 1563 at terminal B	$fr_{1563,B}$	$fr_{1563,A} + 2066.2$ Hz
Index of the tooth of Comb 1563 locking to the ultra-stable laser at terminal B	$N_{63,B}$	967019
Difference of repetition rate of the two Comb 1545	Δfr_{1545}	2582.9 Hz
Difference of repetition rate of the two Comb 1563	Δfr_{1563}	2066.2 Hz
Real frequency of the ultra-stable laser at terminal B	$f_{cw,real,B}$	$[f_{cw,B} - 5.38$ kHz, $f_{cw,B} - 2.39$ kHz]